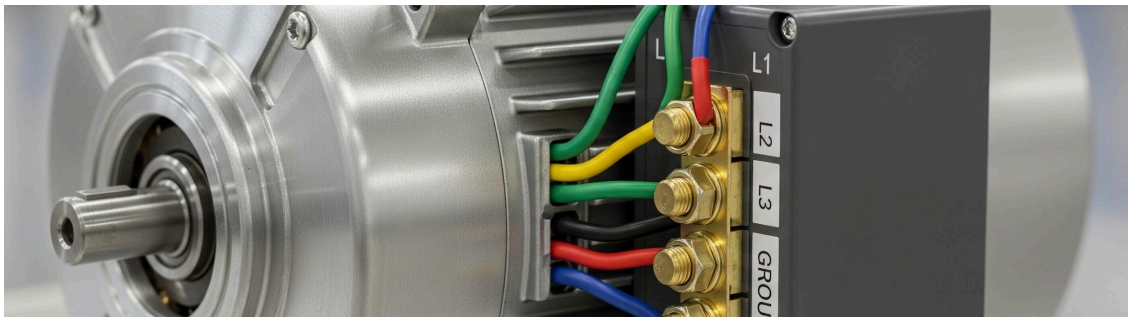

Electrical Machine Data Sheet

Machine Name/Type : Single Phase Induction Motor

Name plate Details

The rating section details the machine's operational capacity and design limits.

Parameter	Value	Unit
Power/Output	1	HP
Voltage	230	V
Amps	7.6	A
Frequency	50	Hz
Phase	1	Phase
Speed	1425	RPM

Precautions

I. Ensure the supply voltage is **230 V AC, single phase** before starting.

II. Do not operate the motor if the capacitor is damaged or disconnected.

III. Use proper protection devices such as MCB and fuse.

IV. Check tightness of terminal connections before switching ON.

V. Do not overload the motor beyond its rated capacity.

VI. Ensure proper ventilation to avoid overheating.

VII. Maintain proper earthing of the motor body.

VIII. Avoid frequent switching ON and OFF.

IX. Keep rotating parts covered during operation.

X. Switch OFF supply immediately if excessive noise, vibration, or overheating occurs.

Working Principle of Single Phase Induction Motor

A single-phase induction motor operates on the principle of **electromagnetic induction**.

When single-phase AC supply is given to the stator winding, it produces a **pulsating magnetic field**, which cannot start the motor on its own.
To create starting torque, an **auxiliary winding with a capacitor** is used to produce a phase difference.

This results in a rotating magnetic field, inducing current in the rotor and producing torque. Once started, the motor continues to run as an induction motor.

Working Characteristics of Single Phase Induction Motor

I. Not self-starting without auxiliary winding.

II. Speed remains nearly constant under normal load.

III. Lower efficiency compared to three-phase motors.

IV. Starting torque depends on the capacitor type used.

V. Simple construction and low maintenance.

VI. Suitable for low and medium power applications.

Applications

1. Ceiling fans
2. Water pumps
3. Refrigerators
4. Washing machines
5. Air coolers
6. Vacuum cleaners
7. Small compressors
8. Mixers and grinders
9. Drilling machines
10. Household appliances